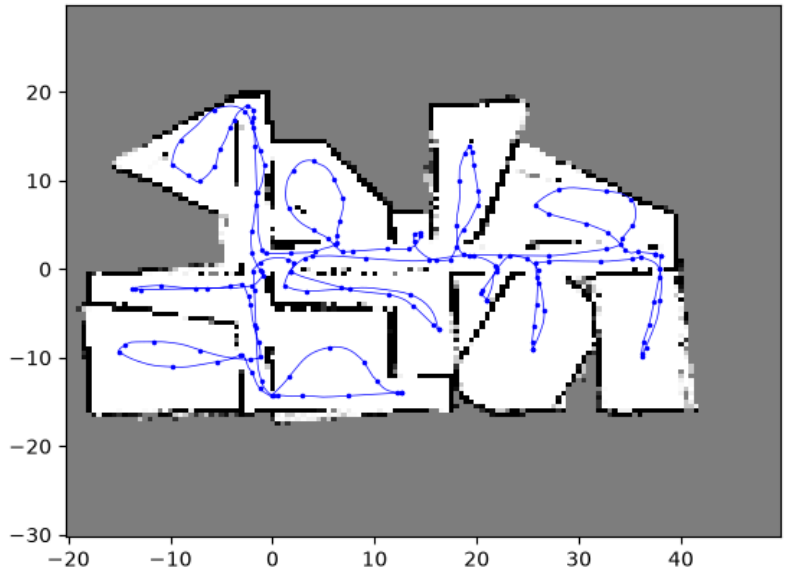
Coding Assignment 2: Occupancy Grid Mapping
Austin J. Erickson

2.2: Implementation

Code in zip file.

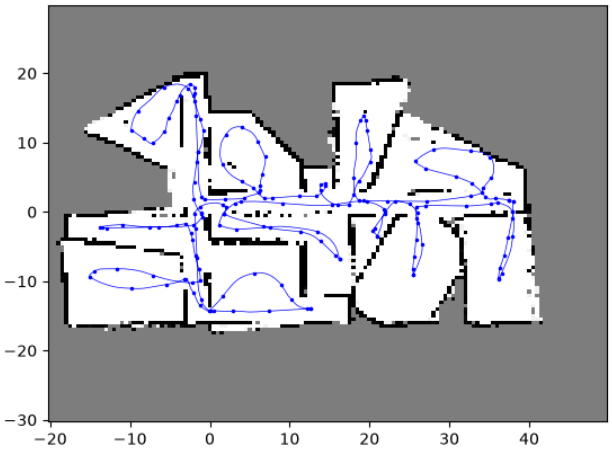
Image from first successful past shown below:

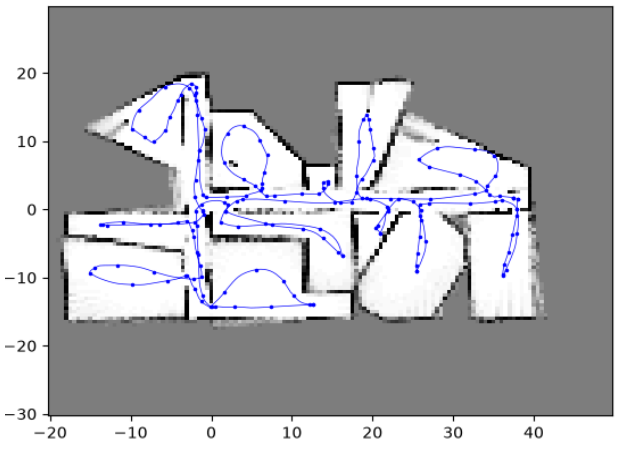


(p_free = 0.1, p_occup = 0.8, p_prior = 0.5)

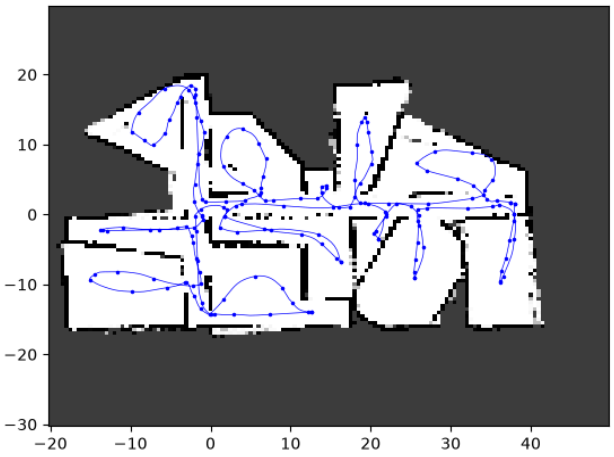
2.3: Laser Range Finder Inverse Sensor Model

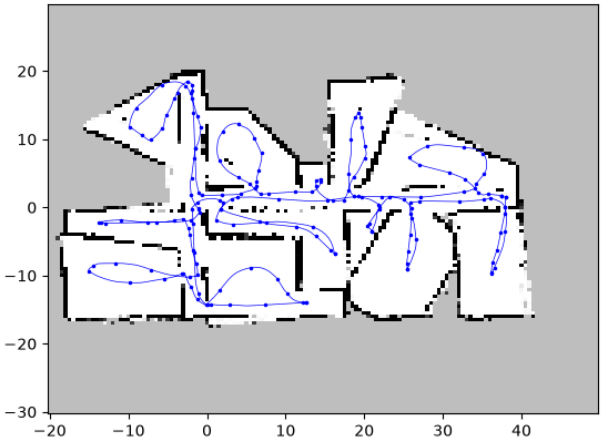
Testing with other p values (multiple different values were tested, here are some of the notable examples):





Left: p_occ = 0.95, p_free = 0.05, p_prior=0.5 Right: p_occ = 0.55, p_free = 0.45, p_prior=0.5





Right: p_occ = 0.9, p_free = 0.1, p_prior=0.75 Left: p_occ = 0.9, p_free = 0.1, p_prior=0.25

These maps show how modulating our values for p can impact our occupancy grid. Changing the prior changes our assumption of how occupied the space is. A higher prior results in more defined rooms and sharper object detection, however it also seems to pick up on noise much sharper as well (e.g. people walking), whereas the lower prior does not define the space as sharply, but is less sensitive to potentially erroneous wall detections. Furthermore, when p_occ and p_free values are closer to the prior it results in the resulting squares (whether more free or occupied) to become less certain, or more hazy, resulting in less definition. When the p_occ and p_free values are further from the p_prior, it results in sharper resolution for the respective squares, though, as with a higher prior, can result in noisy measurements becoming sharper as well.

2.4: A Note on Plotting

A reason for why plotting as you go takes more time consuming is that the values of the squares are constantly being updated with each ray measurement, resulting in potential rapid changing of a single square color in a short amount of time. When plotting this, this results in the display having to change the color of these squares for each change addition of an inverse sensor model to our log odds equation. Without plotting, the log odds formula simply adds the inverse sensor model component to the squares current log odds value, which is relatively quick for most computers.

2.5: Testing

```
(base) austiner99@radlab15:~/classes/localization_and_mapping_f26$ pytest -v -rN --tb=no --no-header
============================================================================= test session starts
==============================================================================
collected 58 items
```

coding_ass_1/byu_mobile_robotic_systems-lab1-prob-review-087e7caa4f9a/test/test_discrete_probability_core.py::TestProbabilityMassFunction::test_throws_when_probabilities_do_not_sum_to_one PASSED [1%]
coding_ass_1/byu_mobile_robotic_systems-lab1-prob-review-087e7caa4f9a/test/test_discrete_probability_core.py::TestProbabilityMassFunction::test_throws_when_passed_negatively_likely_outcomes PASSED [3%]
coding_ass_1/byu_mobile_robotic_systems-lab1-prob-review-087e7caa4f9a/test/test_discrete_probability_core.py::TestJointProbabilityMassFunction::test_throws_when_probabilities_do_not_sum_to_one PASSED [5%]
coding_ass_1/byu_mobile_robotic_systems-lab1-prob-review-087e7caa4f9a/test/test_discrete_probability_core.py::TestJointProbabilityMassFunction::test_throws_when_passed_negatively_likely_outcomes PASSED [6%]
coding_ass_1/byu_mobile_robotic_systems-lab1-prob-review-087e7caa4f9a/test/test_discrete_probability_core.py::TestJointProbabilityMassFunction::test_throws_when_passed_invalid_value PASSED [8%]
coding_ass_1/byu_mobile_robotic_systems-lab1-prob-review-087e7caa4f9a/test/test_part1.py::TestFunctionEvaluatePMF::test_throws_when_probabilities_do_not_sum_to_one PASSED [10%]
coding_ass_1/byu_mobile_robotic_systems-lab1-prob-review-087e7caa4f9a/test/test_part1.py::TestFunctionEvaluatePMF::test_throws_when_passed_negatively_likely_outcomes PASSED [12%]
coding_ass_1/byu_mobile_robotic_systems-lab1-prob-review-087e7caa4f9a/test/test_part1.py::TestFunctionEvaluatePMF::test_correct_for_two_dice_sum PASSED [13%]
coding_ass_1/byu_mobile_robotic_systems-lab1-prob-review-087e7caa4f9a/test/test_part1.py::TestFunctionEvaluatePMF::test_correct_for_two_dice_diff PASSED [15%]
coding_ass_1/byu_mobile_robotic_systems-lab1-prob-review-087e7caa4f9a/test/test_part1.py::TestFunctionEvaluatePMF::test_correct_for_rolling_doubles PASSED [17%]
coding_ass_1/byu_mobile_robotic_systems-lab1-prob-review-087e7caa4f9a/test/test_part1.py::TestFunctionExpectedValue::test_throws_when_probabilities_do_not_sum_to_one PASSED [18%]
coding_ass_1/byu_mobile_robotic_systems-lab1-prob-review-087e7caa4f9a/test/test_part1.py::TestFunctionExpectedValue::test_throws_when_passed_negatively_likely_outcomes PASSED [20%]
coding_ass_1/byu_mobile_robotic_systems-lab1-prob-review-087e7caa4f9a/test/test_part1.py::TestFunctionExpectedValue::test_throws_when_passed_empty_pmf PASSED [22%]
coding_ass_1/byu_mobile_robotic_systems-lab1-prob-review-087e7caa4f9a/test/test_part1.py::TestFunctionExpectedValue::test_correct_for_two_dice_sum PASSED [24%]
coding_ass_1/byu_mobile_robotic_systems-lab1-prob-review-087e7caa4f9a/test/test_part1.py::TestFunctionExpectedValue::test_correct_for_two_dice_diff PASSED [25%]
coding_ass_1/byu_mobile_robotic_systems-lab1-prob-review-087e7caa4f9a/test/test_part1.py::TestFunctionVariance::test_throws_when_probabilities_do_not_sum_to_one PASSED [27%]
coding_ass_1/byu_mobile_robotic_systems-lab1-prob-review-087e7caa4f9a/test/test_part1.py::TestFunctionVariance::test_throws_when_passed_negatively_likely_outcomes PASSED [29%]
coding_ass_1/byu_mobile_robotic_systems-lab1-prob-review-087e7caa4f9a/test/test_part1.py::TestFunctionVariance::test_throws_when_passed_empty_pmf PASSED [31%]
coding_ass_1/byu_mobile_robotic_systems-lab1-prob-review-087e7caa4f9a/test/test_part1.py::TestFunctionVariance::test_correct_for_two_dice_sum PASSED [32%]
coding_ass_1/byu_mobile_robotic_systems-lab1-prob-review-087e7caa4f9a/test/test_part1.py::TestFunctionVariance::test_correct_for_two_dice_diff PASSED [34%]
coding_ass_1/byu_mobile_robotic_systems-lab1-prob-review-087e7caa4f9a/test/test_part2.py::TestFunctionEvaluateJointPMF::test_throws_when_probabilities_do_not_sum_to_one PASSED [36%]
coding_ass_1/byu_mobile_robotic_systems-lab1-prob-review-087e7caa4f9a/test/test_part2.py::TestFunctionEvaluateJointPMF::test_throws_when_passed_negatively_likely_outcomes PASSED [37%]
coding_ass_1/byu_mobile_robotic_systems-lab1-prob-review-087e7caa4f9a/test/test_part2.py::TestFunctionEvaluateJointPMF::test_correct_for_two_dice_sum_and_two_dice_difference PASSED [39%]
coding_ass_1/byu_mobile_robotic_systems-lab1-prob-review-087e7caa4f9a/test/test_part2.py::TestFunctionEvaluateJointPMF::test_correct_for_two_dice_sum_and_two_dice_sum_plus_one PASSED [41%]
coding_ass_1/byu_mobile_robotic_systems-lab1-prob-review-087e7caa4f9a/test/test_part2.py::TestFunctionEvaluateJointPMF::test_correct_for_two_dice_sum_and_weighted_two_dice_sum PASSED [43%]
coding_ass_1/byu_mobile_robotic_systems-lab1-prob-review-087e7caa4f9a/test/test_part2.py::TestFunctionExpectedValue::test_throws_when_probabilities_do_not_sum_to_one PASSED [44%]
coding_ass_1/byu_mobile_robotic_systems-lab1-prob-review-087e7caa4f9a/test/test_part2.py::TestFunctionExpectedValue::test_throws_when_passed_negatively_likely_outcomes PASSED [46%]
coding_ass_1/byu_mobile_robotic_systems-lab1-prob-review-087e7caa4f9a/test/test_part2.py::TestFunctionExpectedValue::test_throws_when_passed_empty_pmf PASSED [48%]
coding_ass_1/byu_mobile_robotic_systems-lab1-prob-review-087e7caa4f9a/test/test_part2.py::TestFunctionExpectedValue::test_correct_for_two_dice_sum_and_two_dice_difference PASSED [50%]
coding_ass_1/byu_mobile_robotic_systems-lab1-prob-review-087e7caa4f9a/test/test_part2.py::TestFunctionExpectedValue::test_correct_for_two_dice_sum_and_two_dice_sum_plus_one PASSED [51%]

coding_ass_1/byu_mobile_robotic_systems-lab1-prob-review-087e7caa4f9a/test/test_part2.py::TestFunctionExpectedValue::test_correct_for_two_dice_sum_and_weighted_two_dice_sum PASSED [53%]
coding_ass_1/byu_mobile_robotic_systems-lab1-prob-review-087e7caa4f9a/test/test_part2.py::TestFunctionCovariance::test_throws_when_probabilities_do_not_sum_to_one PASSED [55%]
coding_ass_1/byu_mobile_robotic_systems-lab1-prob-review-087e7caa4f9a/test/test_part2.py::TestFunctionCovariance::test_throws_when_passed_negatively_likely_outcomes PASSED [56%]
coding_ass_1/byu_mobile_robotic_systems-lab1-prob-review-087e7caa4f9a/test/test_part2.py::TestFunctionCovariance::test_throws_when_passed_empty_pmf PASSED [58%]
coding_ass_1/byu_mobile_robotic_systems-lab1-prob-review-087e7caa4f9a/test/test_part2.py::TestFunctionCovariance::test_correct_for_two_dice_sum_and_two_dice_difference PASSED [60%]
coding_ass_1/byu_mobile_robotic_systems-lab1-prob-review-087e7caa4f9a/test/test_part2.py::TestFunctionCovariance::test_correct_for_two_dice_sum_and_two_dice_sum_plus_one PASSED [62%]
coding_ass_1/byu_mobile_robotic_systems-lab1-prob-review-087e7caa4f9a/test/test_part2.py::TestFunctionCovariance::test_correct_for_two_dice_sum_and_weighted_two_dice_sum PASSED [63%]
coding_ass_1/byu_mobile_robotic_systems-lab1-prob-review-087e7caa4f9a/test/test_part3.py::TestFunctionMarginalizeOut::test_throws_when_invalid_indicator_passed_in PASSED [65%]
coding_ass_1/byu_mobile_robotic_systems-lab1-prob-review-087e7caa4f9a/test/test_part3.py::TestFunctionMarginalizeOut::test_throws_when_invalid_joint_pmf_received PASSED [67%]
coding_ass_1/byu_mobile_robotic_systems-lab1-prob-review-087e7caa4f9a/test/test_part3.py::TestFunctionMarginalizeOut::test_correct_for_marginalize_out_difference_from_sum_and_difference PASSED [68%]
coding_ass_1/byu_mobile_robotic_systems-lab1-prob-review-087e7caa4f9a/test/test_part3.py::TestFunctionMarginalizeOut::test_correct_for_marginalize_out_sum_from_sum_and_difference PASSED [70%]
coding_ass_1/byu_mobile_robotic_systems-lab1-prob-review-087e7caa4f9a/test/test_part3.py::TestFunctionConditionAgainst::test_throws_when_invalid_indicator_passed_in PASSED [72%]
coding_ass_1/byu_mobile_robotic_systems-lab1-prob-review-087e7caa4f9a/test/test_part3.py::TestFunctionConditionAgainst::test_throws_when_invalid_joint_pmf_received PASSED [74%]
coding_ass_1/byu_mobile_robotic_systems-lab1-prob-review-087e7caa4f9a/test/test_part3.py::TestFunctionConditionAgainst::test_correct_for_condition_against_difference_from_sum_and_difference PASSED [75%]
coding_ass_1/byu_mobile_robotic_systems-lab1-prob-review-087e7caa4f9a/test/test_part3.py::TestFunctionConditionAgainst::test_correct_for_condition_against_sum_from_sum_and_difference PASSED [77%]
coding_ass_2/lab_2/tests/test_cells_to_update.py::test[-90] PASSED [79%]
coding_ass_2/lab_2/tests/test_cells_to_update.py::test[0] PASSED [81%]
coding_ass_2/lab_2/tests/test_cells_to_update.py::test[90] PASSED [82%]
coding_ass_2/lab_2/tests/test_cells_to_update.py::test[180] PASSED [84%]
coding_ass_2/lab_2/tests/test_integrate_laser.py::test PASSED [86%]
coding_ass_2/lab_2/tests/test_inverse_sensor.py::test_free PASSED [87%]
coding_ass_2/lab_2/tests/test_inverse_sensor.py::test_occup PASSED [89%]
coding_ass_2/lab_2/tests/test_inverse_sensor.py::test_prior PASSED [91%]
coding_ass_2/lab_2/tests/test_update_cell.py::test[0] PASSED [93%]
coding_ass_2/lab_2/tests/test_update_cell.py::test[1] PASSED [94%]
coding_ass_2/lab_2/tests/test_update_cell.py::test[2] PASSED [96%]
coding_ass_2/lab_2/tests/test_update_cell.py::test[3] PASSED [98%]
coding_ass_2/lab_2/tests/test_update_cell.py::test[4] PASSED [100%]

============================== 58 passed in 0.33s ==============================

2.6: Wrap Up:

Info in submission